

ISC Class XII Maths — Chapter 2

Inverse Trigonometric Functions — Formula Sheet

A. Domains & Principal Value Branches

$$\sin^{-1} x : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \cos^{-1} x : [0, \pi] \quad \tan^{-1} x : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\cot^{-1} x : (0, \pi) \quad \sec^{-1} x : [0, \pi] \setminus \left\{\frac{\pi}{2}\right\} \quad \operatorname{cosec}^{-1} x : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$$

Input domains: $\sin^{-1}, \cos^{-1}$ need $|x| \leq 1$; $\tan^{-1}, \cot^{-1}$ take all $x \in \mathbb{R}$; $\sec^{-1}, \operatorname{cosec}^{-1}$ need $|x| \geq 1$.

B. Inverse-Composition Identities

$f(f^{-1}(x)) = x$ whenever f^{-1} is defined at x (e.g. $\sin(\sin^{-1} x) = x, |x| \leq 1$).

$f^{-1}(f(\theta)) = \theta$ **only if** θ already lies in f^{-1} 's principal branch (Section A) — otherwise reduce θ into range first.

Tip: For $f^{-1}(f(\theta))$ with θ outside range: rewrite using a supplementary/periodic identity (e.g. $\sin(\pi - \theta) = \sin \theta$, $\cos(2\pi - \theta) = \cos \theta$, $\tan(\theta - \pi) = \tan \theta$) until the reduced angle sits inside the branch.

C. Negative-Argument Identities

Odd: $\sin^{-1}(-x) = -\sin^{-1} x$ $\tan^{-1}(-x) = -\tan^{-1} x$
 $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$

Supplementary: $\cos^{-1}(-x) = \pi - \cos^{-1} x$ $\cot^{-1}(-x) = \pi - \cot^{-1} x$
 $\sec^{-1}(-x) = \pi - \sec^{-1} x$

D. Complementary-Angle (Co-function) Pairs

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, |x| \leq 1 \quad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}, x \in \mathbb{R}$$

$$\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}, |x| \geq 1$$

E. Reciprocal / Conversion Identities

$$\sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1} x, |x| \geq 1 \quad \cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1} x, |x| \geq 1$$

$$\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1} x, x > 0; \text{ for } x < 0: \tan^{-1}\left(\frac{1}{x}\right) = -\pi + \cot^{-1} x$$

F. Sum / Difference — Tangent Inverse

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right), xy < 1$$

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1}\left(\frac{x-y}{1+xy}\right), xy > -1$$

Tip: If $x, y > 0$ and $xy > 1$: add π to the RHS. If $x, y < 0$ and $xy > 1$: subtract π . **Triple sum:** $\tan(A+B+C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$ — use when three inverse-tangent terms add to a known angle.

G. Sum Formulas — Sine & Cosine Inverse

$$\sin^{-1} x + \sin^{-1} y = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$$

$$\cos^{-1} x + \cos^{-1} y = \cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right)$$

Tip: If the result falls outside the natural principal range, adjust by $\pi - (\cdot)$ or add π as the sign/quadrant of x, y demands.

H. Double-Angle ($2x$) Forms

$$2 \tan^{-1} x = \sin^{-1}\left(\frac{2x}{1+x^2}\right), |x| \leq 1$$

$$2 \tan^{-1} x = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right), x \geq 0$$

$$2 \tan^{-1} x = \tan^{-1}\left(\frac{2x}{1-x^2}\right), -1 < x < 1$$

Tip: These come from putting $x = \tan \theta$ into $\sin 2\theta, \cos 2\theta, \tan 2\theta$ — spot them whenever an expression contains $\frac{2x}{1+x^2}, \frac{1-x^2}{1+x^2}$, or $\frac{2x}{1-x^2}$.

Worked Pattern: Right-Triangle Conversion

To convert $\theta = \sin^{-1}\left(\frac{p}{h}\right)$ into another ratio:

1. Draw a right triangle: opposite = p , hypotenuse = h .
2. Adjacent = $\sqrt{h^2 - p^2}$ (Pythagoras).
3. Read off the needed ratio (e.g. $\tan \theta = \frac{p}{\sqrt{h^2 - p^2}}$).

Works the same starting from $\cos^{-1}$ or $\tan^{-1}$ — always label opposite/adjacent/hypotenuse first, then Pythagoras, then read the ratio.

Worked Pattern: Weierstrass (Half-Angle) Substitution

For expressions mixing $\sin \theta$ and $\cos \theta$ inside a $\tan^{-1}$:

1. Let $t = \tan \frac{\theta}{2}$.
2. Convert: $\sin \theta = \frac{2t}{1+t^2}$, $\cos \theta = \frac{1-t^2}{1+t^2}$, $\tan \theta = \frac{2t}{1-t^2}$.
3. Substitute, simplify the resulting expression in t .
4. Back-substitute $t = \tan \frac{\theta}{2}$ in the final answer.

K. Half-Angle Trig Identities (for simplification)

$$1 - \cos x = 2 \sin^2 \frac{x}{2} \quad 1 + \cos x = 2 \cos^2 \frac{x}{2}$$

$$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2} \quad \tan\left(\frac{\pi}{4} - x\right) = \frac{1 - \tan x}{1 + \tan x}$$

Tip: Standard substitutions for $\tan^{-1}(f(\sin x, \cos x))$ -type simplification: $x = \sin \theta \Rightarrow \sqrt{1-x^2} = \cos \theta$; $x = \tan \theta \Rightarrow \sqrt{1+x^2} = \sec \theta$; $x = \cos \theta \Rightarrow \sqrt{1-x^2} = \sin \theta$; $x = a \sin \theta \Rightarrow \sqrt{a^2 - x^2} = a \cos \theta$.

M. Common Traps to Avoid

- $\cos^{-1}(-x) \neq -\cos^{-1} x$ — it equals $\pi - \cos^{-1} x$ (range is $[0, \pi]$, never negative).
- $f^{-1}(f(\theta)) = \theta$ only when θ is already inside the principal branch — always check first.
- In $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$, forgetting the $\pm\pi$ correction when $xy > 1$ silently flips the sign of the answer.
- When solving equations, always verify each root against the *original* domain — squaring or combining steps can introduce extraneous roots.
- $\sec^{-1} x$ and $\operatorname{cosec}^{-1} x$ are undefined for $-1 < x < 1$; don't apply the reciprocal identities there.

L. Quick Recognition Guide

Pattern seen	Use
$f^{-1}(f(\theta))$, θ outside branch	Sec. B — reduce θ first
$\cos(\sin^{-1} x)$, $\tan(\cos^{-1} x)$, etc.	Right-Triangle Method
$f(-x)$ inside inverse trig	Sec. C odd/supplementary
Two inverse terms	Sec. D (co-function) or F/G summing
$\frac{1}{x}$ inside inverse trig	Sec. E reciprocal identity
$xy > 1$ in $\tan^{-1} x + \tan^{-1} y$	Sec. F, add/subtract π
Contains $\frac{2x}{1+x^2}$, $\frac{1-x^2}{1+x^2}$, $\sqrt{\frac{1-\cos x}{1+\cos x}}$ -type	Sec. H double-angle form Sec. K half-angle $\rightarrow$ Weierstrass
Solve for x : sum of two inverse terms = const	Combine (F/G), then invert
Three inverse-tangent terms sum to known angle	Sec. F triple-tangent identity
Given $x-y-xy$ or similar from a difference equation	Sec. F, isolate via $\tan$ of both sides
"Find the max/min value of $(\cdot)^{-1}x)^k$ "	Sec. A — use endpoint of principal range

Exam Tip: Always state the principal-value range you are working in *before* quoting a numeric answer — ISC markers award a step for this, and it is the single most common one-mark deduction in this chapter.